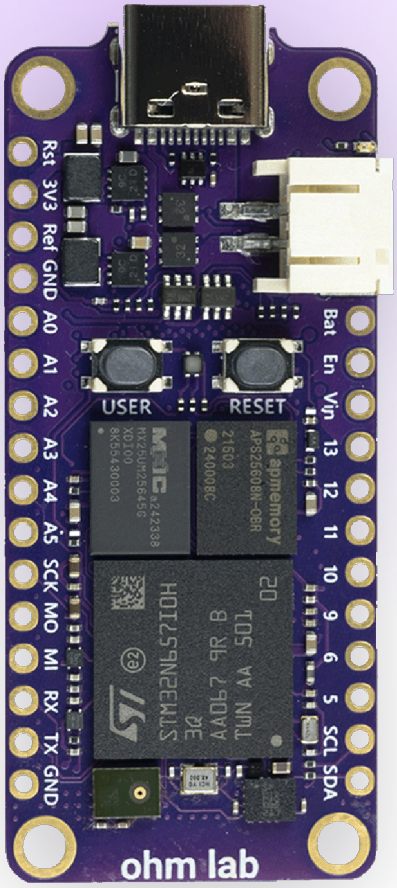
Neuro N6

Datasheet



Arduino-compatible edge AI development board built on the STM32N657 microcontroller. It combines an 800 MHz Arm Cortex-M55 core with the ST Neural-ART neural processing unit, 32 MB of octal SPI flash and a separate 64 MB octal SPI PSRAM, USB-C power with Li-ion charging, and two board-to-board connectors for the Neuro Vision camera, thermal and display expansion boards.

600 GOPS
ST Neural-ART accelerator

800 MHz
Arm Cortex-M55

32 MB
Octal SPI flash

64 MB
Octal SPI PSRAM

PART NUMBER	HARDWARE REVISION	DOCUMENT	DATE
OL-DEV-N6-001	Rev C	OL-DEV-N6-001 Rev E	25 October 2026

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1 Overview

The Neuro N6 is an edge AI development board in the Adafruit Feather form factor. It pairs the STM32N657 microcontroller with external flash and PSRAM, on-board motion and audio sensors, a Li-ion power path, and connectors for the Neuro Vision expansion boards.

- STM32N657I0H3Q microcontroller: Arm Cortex-M55 at 800 MHz with Helium vector extension, ST Neural-ART accelerator (600 GOPS), 4.2 MB contiguous SRAM, H.264 and MJPEG video encoder, Neo-Chrom 2.5D GPU.
- 32 MB octal SPI flash (MX25UM25645G) and a separate 64 MB octal SPI PSRAM (APS51208N), each with its own chip select on the 1.8 V XSPI memory port (up to 200 MHz DTR).
- Power from USB-C (5 V) or a single-cell Li-ion battery, with ideal-diode OR-ing and a linear charger programmed to 500 mA.
- On-board sensors: LSM6DSL 6-axis IMU, MMC5603NJ 3-axis magnetometer and MP34DT05-A PDM microphone.
- Feather-format headers: UART4, SPI2, I2C2, six ADC inputs (1.8 V full scale), seven digital I/O, five timer PWM outputs, reset and enable inputs, and the USB (VBUS), battery, 3.3 V and 1.8 V analog reference pins. The header pins can be reconfigured as an RMII interface to an external Ethernet PHY.
- One 40-pin and one 30-pin board-to-board connector carrying 2-lane MIPI CSI-2, 4-bit SDMMC, I2C1, RGB565 LTDC, 4-wire USART6, SAI1 audio, SWD and power.
- USB 2.0 high speed (480 Mbit/s) device port with the STM32N6 internal PHY, used for power, the Ohm Lab bootloader and application data.
- Supported by the Neuro N6 Arduino core, with pre-built examples for camera capture and neural network inference.

32 MB	64 MB	480 Mbit/s	500 mA	~750 mW
Octal SPI flash	Octal SPI PSRAM	USB 2.0 high speed	Li-ion charger	Typical power consumption

2 Introduction

The Neuro N6 development board is an accessible edge AI platform with an on-board neural processing unit. It allows edge AI applications to be developed without prior experience of artificial intelligence or embedded systems, and its capabilities are extended by the Neuro Vision range of expansion boards.

Because it is built on a microcontroller rather than a Linux processor, the Neuro N6 starts instantly, wakes from low-power sleep in milliseconds and draws little power, so it suits battery-powered devices that must react quickly and run for a long time: smart cameras that wake on an event, vision and audio sensing, wearables and remote sensor nodes, with neural network inference running on the board without an operating system.

The Neuro N6 Arduino core, available from neuron6.com, provides a programming interface through the Arduino IDE together with example applications that run on the board immediately after installation. The board is supplied with the Ohm Lab bootloader and a placeholder program only: install the board support package for the Arduino IDE and follow the quick-start guide at neuron6.com.

This document describes the features, electrical characteristics and mechanical specification of the Neuro N6. Operating and programming instructions are given in the tutorials at wiki.neuron6.com. The Neuro Vision expansion boards are described in their own datasheets.

For detailed information on the STM32N657 microcontroller, refer to the STMicroelectronics datasheet DS14791 and reference manual RM0484.

2.1 Ordering information

PRODUCT	PART NUMBER	HARDWARE REVISION	NOTES
Neuro N6	`OL-DEV-N6-001`	Rev C	Supplied without headers fitted

Table 1. Ordering information. Quote the part number and hardware revision when ordering and when contacting Ohm Lab for support.

The hardware revision is printed on the underside of the board next to the part number (`OL-DEV-N6-001 : C`).

2.2 Related products

- Neuro Vision OV5640 (OL-DEV-N6-002)
- Neuro Vision OV5640-W (OL-DEV-N6-003)
- Neuro Vision ST Cam (OL-DEV-N6-004)
- Neuro Vision Thermal (OL-DEV-N6-005)
- Neuro Vision TFT (OL-DEV-N6-006)
- Neuro Vision TFT-W (OL-DEV-N6-007)
- Neuro N6 Enclosure (OL-DEV-N6-009)
- Neuro N6 Debug Adapter (OL-DEV-N6-010)
- B-CAMS-IMX Adapter (OL-DEV-N6-011)

3 Features

FEATURE	DESCRIPTION
STM32N657I0H3Q microcontroller	
Arm Cortex-M55 core	800 MHz, 32 KB instruction cache, 32 KB data cache, Arm M-profile vector extension (Helium), TrustZone, MPU
ST Neural-ART accelerator	1 GHz neural processing unit, 600 GOPS, 288 MAC per cycle, dedicated stream processing engine, real-time weight decryption and decompression
Memory	4.2 MB contiguous SRAM, 128 KB data TCM and 64 KB instruction TCM with ECC
Graphics and video	Neo-Chrom 2.5D GPU (scaling, rotation, alpha blending, texture mapping), Chrom-ART DMA2D, hardware JPEG codec, H.264 and MJPEG encoder, LCD-TFT controller up to XGA
Clocks	48 MHz crystal (HSE) and 32.768 kHz crystal (LSE) for the real-time clock
Package	VFBGA178, 0.8 mm pitch, 12 x 12 mm
External memory	
MX25UM25645G flash	256 Mbit (32 MB) octal SPI NOR flash, 1.8 V, up to 200 MHz DTR (400 MB/s)
APS51208N-OBR-BD PSRAM	512 Mbit (64 MB) octal SPI pseudo-static RAM, 1.8 V, up to 200 MHz DTR
On-board sensors	
LSM6DSL inertial measurement unit	3-axis accelerometer (±2/4/8/16 g), 3-axis gyroscope (±125/245/500/1000/2000 dps), embedded temperature sensor, I2C1 address 0x6A, two interrupt outputs
MMC5603NJ magnetometer	3-axis, ±30 G range, 0.0625 mG resolution, I2C1 address 0x30
MP34DT05-A microphone	Omnidirectional MEMS digital microphone, PDM output to the STM32N6 digital filter (MDF1); 16-bit mono at 32.55 kHz with the Arduino core
Interfaces	
Feather headers	UART4, SPI2, I2C2, six ADC inputs (A0 to A5, 1.8 V full scale), seven digital I/O (D5, D6, D9, D10, D11, D12, D13), five timer PWM outputs (D5, D12, D13, A4, A5), RST and EN inputs, USB VBUS (Vin pin), battery, 3.3 V and 1.8 V analog reference (AREF) pins. Alternate RMII Ethernet function for an external PHY.
40-pin board-to-board connector	2-lane MIPI CSI-2 with D-PHY (up to 5 MP camera modules), SDMMC1 4-bit (also SPI3), I2C1, LTDC (13 data bits, DCLK, VSYNC), four GPIO
30-pin board-to-board connector	USART6 with RTS and CTS, SAI1 block B audio, LTDC (G2, G4, R5, DE, HSYNC), SWDIO, SWCLK, NRST, five GPIO, VIN, 3.3 V and 1.8 V
USB-C	USB 2.0 high speed device (480 Mbit/s) using the internal PHY; 5 V power input
Battery connector	JST PH 2-pin, single-cell 3.7 V Li-ion or LiPo with integral protection circuit, charged from USB-C at up to 500 mA
Debug connector	6-pin 0.8 mm: SWDIO, SWCLK, NRST, BOOT1, 3.3 V, GND; Neuro N6 Debug Adapter (OL-DEV-N6-010) for standard debug probes

FEATURE	DESCRIPTION
User interface	
LEDs	User RGB LED driven by timer PWM outputs; red charge status LED
Buttons	RESET button and USER button

Table 2. Neuro N6 features.

4 Mechanical

The Neuro N6 follows the Adafruit Feather outline, extended by a 12.70 mm wide, 1.27 mm deep tab at the USB-C end. The four mounting holes are on the Feather 45.72 x 17.78 mm pattern. All dimensions are in millimetres and are taken from the Rev C PCB design data and the 3D model.

PCB outline	52.07 x 22.86 mm (50.80 mm Feather body plus 1.27 mm tab), corner radius 2.54 mm
Overall length including USB-C receptacle	53.04 mm
Overall height	8.85 mm: 5.50 mm above the top surface (battery connector), 1.60 mm PCB, 1.75 mm below the PCB (debug connector)
Board-to-board plugs	1.14 mm below the PCB; 3.5 mm mating height with the expansion board receptacles
PCB	1.6 mm, 8 copper layers
Mounting holes	4 x Ø2.54 mm plated, 45.72 x 17.78 mm pattern, 2.54 mm from the Feather body edges, connected to GND
Header holes	28 x Ø1.0 mm plated, 2.54 mm pitch, rows 20.32 mm apart
Mass	About 6.7 g, without headers

Table 3. Mechanical summary.

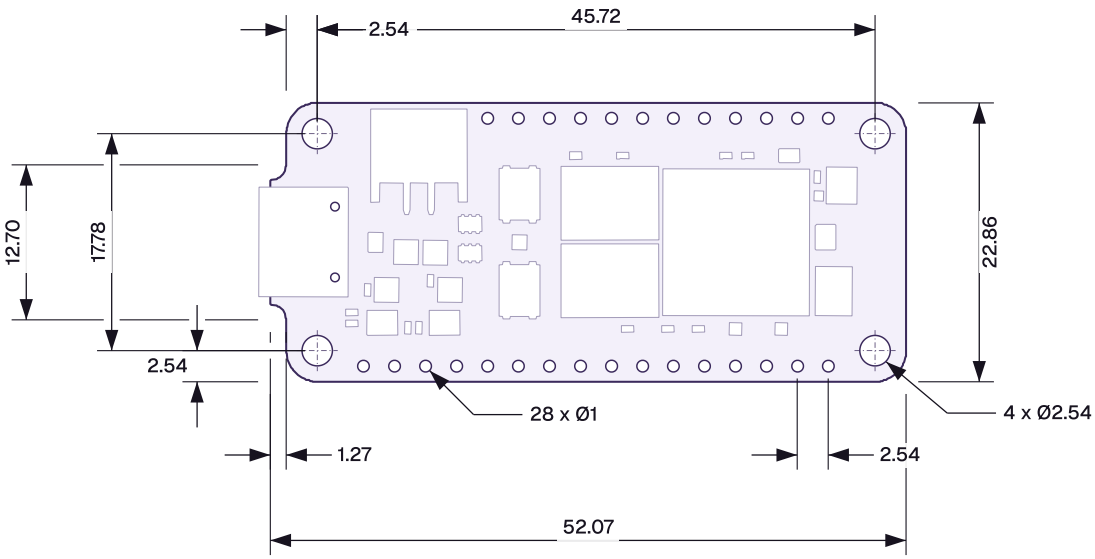


Figure 1. Neuro N6 top view. Component outlines are taken from the 3D model; dimensions are nominal design values in millimetres.

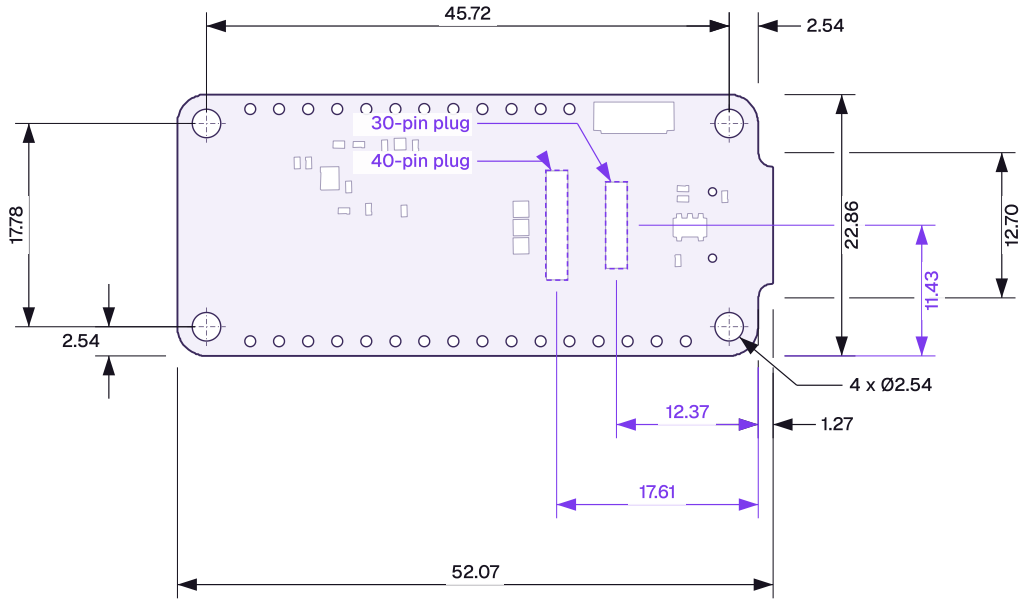


Figure 2. Neuro N6 bottom view (mirrored, as seen from below). Plug positions are given from the Feather body edge at the USB-C end.

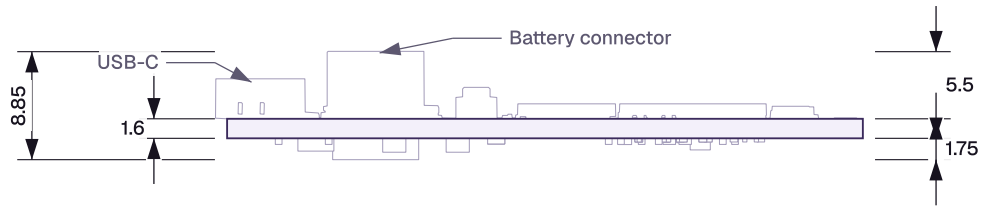


Figure 3. Neuro N6 side view.

5 Block diagram

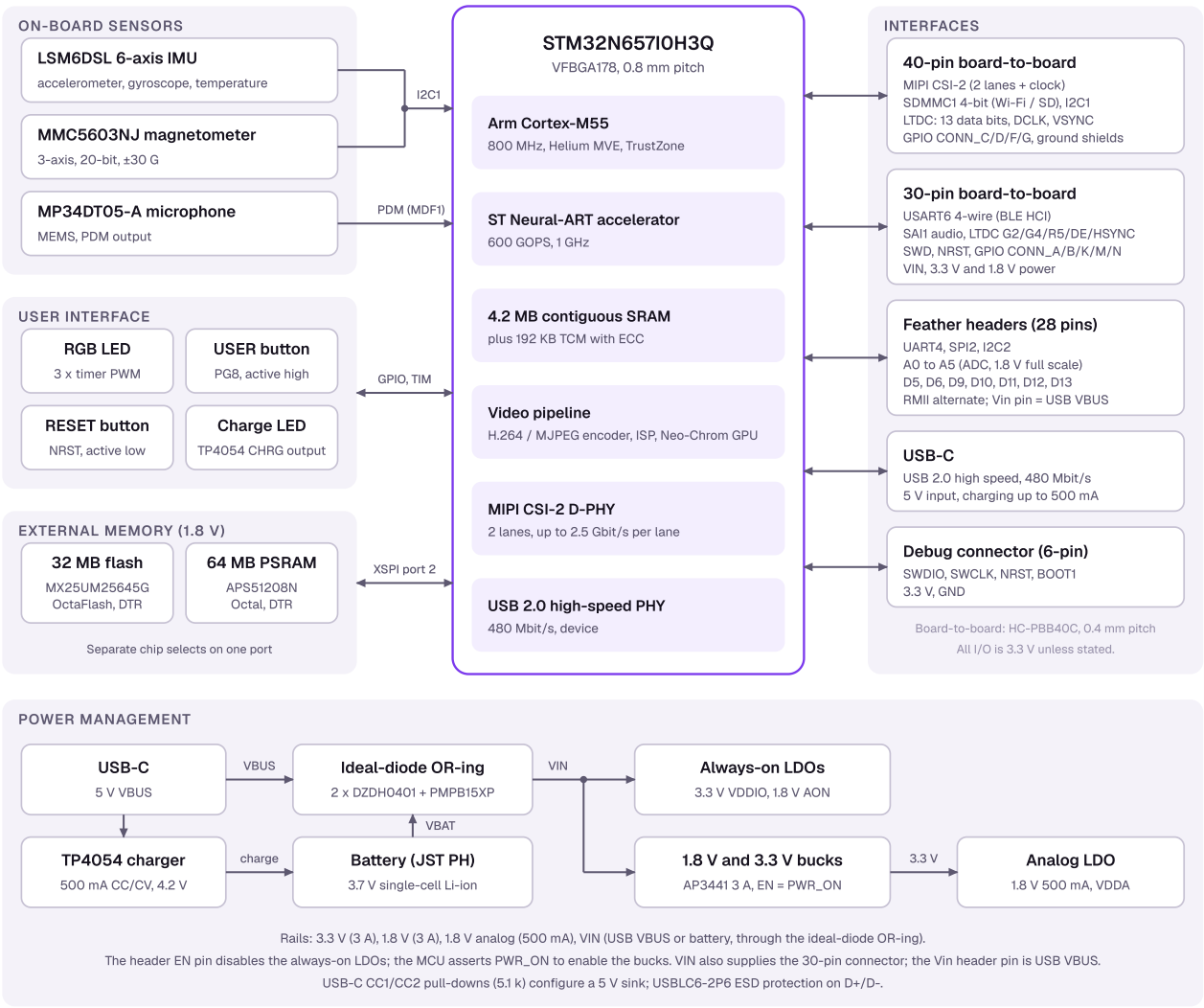


Figure 4. Neuro N6 system block diagram.

6 Hardware interfaces

6.1 Header pins

The 28 header positions follow the Adafruit Feather pin assignment. Headers are not fitted as supplied. All header signals are 3.3 V logic. The analog inputs A0 to A5 are routed to STM32N6 ADC channels: as digital I/O these pins accept 3.3 V, but the ADC full-scale input is 1.8 V, set by the 1.8 V VREF+ supply that is also available on the AREF pin.

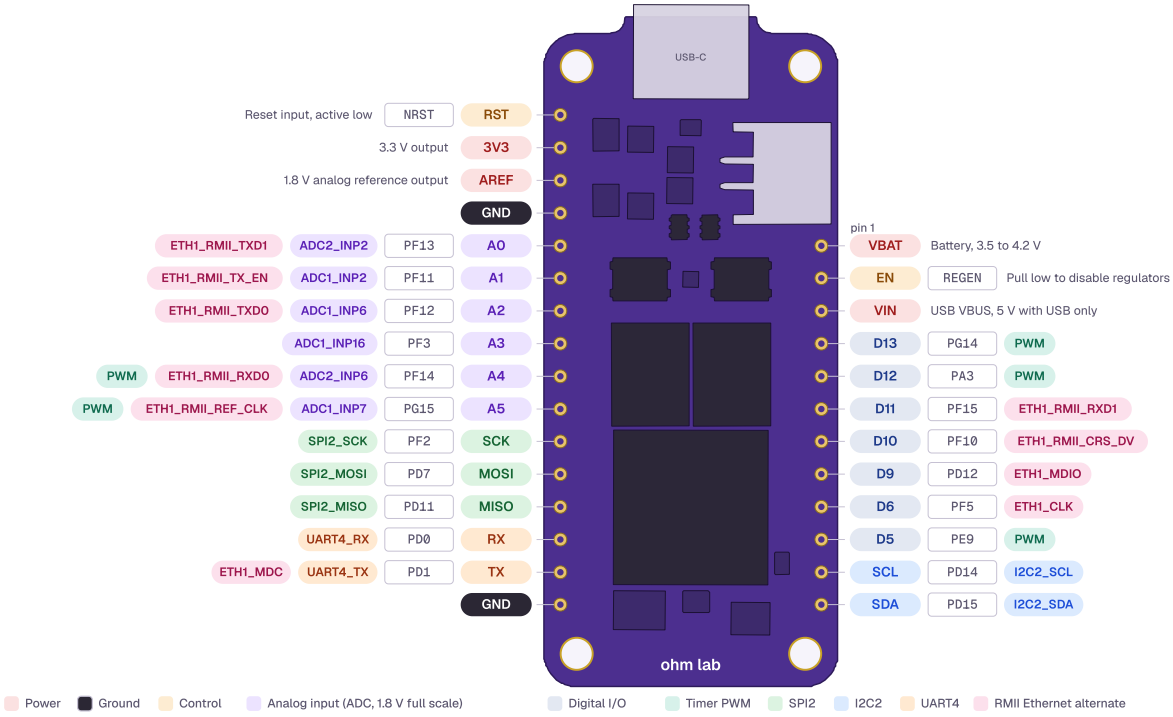


Figure 5. Neuro N6 header pin assignment (top view). MCU pin names are STM32N657 port pins; the RMII functions apply when the header pins are configured for an external Ethernet PHY.

> **Note** Feather compatibility: the pin positions follow the Adafruit Feather layout. The pin printed Vin carries USB VBUS, as the USB pin of a Feather does: it has 5 V only while USB is connected and is not powered from the battery. AREF is an output tied to the 1.8 V analog rail and must not be driven. The ADC full-scale input is 1.8 V, so analog sensors made for 3.3 V Feather boards need a divider.

6.2 Header pin table

PIN	NAME	MCU PIN	GROUP	FUNCTION
1	VBAT		Power	Battery terminal, 3.5 to 4.2 V. Output when a battery is fitted, input to the charger from USB-C.
2	EN	REGEN	Control	Regulator enable. Pulled up to VIN through 100 kΩ; pull low to switch off the always-on regulators and the board.
3	VIN		Power	USB VBUS: 5 V from the USB-C port, before the ideal-diode OR-ing. Present only while USB is connected; not powered from the battery. Do not connect another 5 V supply while USB is connected.
4	D13	PG14	Digital I/O	TIM8_CH4 timer PWM
5	D12	PA3	Digital I/O	TIM16_CH1 timer PWM
6	D11	PF15	Digital I/O	ETH1_RMII_RXD1 in Ethernet mode
7	D10	PF10	Digital I/O	ETH1_RMII_CRS_DV in Ethernet mode
8	D9	PD12	Digital I/O	ETH1_MDIO in Ethernet mode
9	D6	PF5	Digital I/O	ETH1_CLK in Ethernet mode
10	D5	PE9	Digital I/O	TIM1_CH1 timer PWM
11	SCL	PD14	I2C2	I2C2_SCL, 4.7 kΩ pull-up to 3.3 V
12	SDA	PD15	I2C2	I2C2_SDA, 4.7 kΩ pull-up to 3.3 V
13	GND		Ground	
14	TX	PD1	UART4	UART4_TX; ETH1_MDC in Ethernet mode
15	RX	PD0	UART4	UART4_RX
16	MISO	PD11	SPI2	SPI2_MISO
17	MOSI	PD7	SPI2	SPI2_MOSI
18	SCK	PF2	SPI2	SPI2_SCK
19	A5	PG15	Analog	ADC1_INP7 / ADC2_INP7; TIM1_CH4 PWM; ETH1_RMII_REF_CLK in Ethernet mode
20	A4	PF14	Analog	ADC2_INP6; TIM2_CH2 PWM; ETH1_RMII_RXD0 in Ethernet mode
21	A3	PF3	Analog	ADC1_INP16
22	A2	PF12	Analog	ADC1_INP6; ETH1_RMII_TXD0 in Ethernet mode
23	A1	PF11	Analog	ADC1_INP2; ETH1_RMII_TX_EN in Ethernet mode
24	A0	PF13	Analog	ADC2_INP2; ETH1_RMII_TXD1 in Ethernet mode
25	GND		Ground	
26	AREF		Power	1.8 V analog reference output (VDDA_1V8), also the ADC reference. Output only, for reference use: keep the load below 10 mA
27	3V3		Power	3.3 V rail output
28	RST	NRST	Control	System reset, active low. Connected to the RESET button and the debug connector.

Table 4. Header pin assignment. Pin numbers count clockwise from VBAT, as shown in the pin diagram.

6.3 Buttons

- **RESET**: pulls NRST low to reset the microcontroller and, through the expansion connectors, the expansion boards. Two presses within 900 ms enter the Ohm Lab bootloader (see the boot modes section).
- **USER**: connects PG8 to 3.3 V. The pin is held low by a 10 kΩ resistor, so the button reads high when pressed. The pin is protected by an ESDZL5-1F4 diode and is unused by default.

6.4 LEDs

- **RGB LED**: XL-B1212RGBA common-anode LED (anode to 3.3 V) driven from three timer outputs (red PE11, green PD6, blue PC1) through series resistors, so the outputs are active low. It is fully user programmable. During bootloader mode the LED pulses red.
- **Charge LED** (red): driven by the TP4054 CHRG output. On while the battery is charging; off when charging is complete or when no USB power is present. With USB power and no battery connected the charger cycles and the LED flickers rapidly.

6.5 Sensors

The three on-board sensors are described in the manufacturers' datasheets. Their connections on the Neuro N6 are:

SENSOR	PART	INTERFACE	ADDRESS / SIGNALS
Inertial measurement unit	LSM6DSL	I2C1 (shared with the 40-pin connector)	`0x6A` (SA0 low); INT1 on PE14, INT2 on PE15
Magnetometer	MMC5603NJ	I2C1 (shared with the 40-pin connector)	`0x30`
Microphone	MP34DT05-A	PDM to MDF1	Clock PE2 (MDF1_CCK0), data PB2 (MDF1_SDI1), L/R select tied low

Table 5. On-board sensor connections.

6.6 Connectors

CONNECTOR	PART NUMBER	MATING PART	NOTES
USB-C receptacle	`TYPE-C-31-M-12`	USB-C plug	16-pin, USB 2.0 pins only; CC1 and CC2 pulled down with 5.1 kΩ (5 V sink)
Battery connector	`S2B-PH-SM4-TB(LF)(SN)`	`PHR-2`	JST PH, 2 mm pitch, right angle. Pin 1 VBAT, pin 2 GND
Debug connector	`M0800RS-06-GN`	`M0800HI-06-GN`	6-pin, 0.8 mm pitch, underside of the board. Neuro N6 Debug Adapter (OL-DEV-N6-010) available
30-pin board-to-board plug	`HC-PBB40C-30DP-0.4V-02`	`HC-PBB40C-30DS-0.4V-3.5-02`	0.4 mm pitch, 3.5 mm mating height, 0.3 A per contact
40-pin board-to-board plug	`HC-PBB40C-40DP-0.4V-02`	`HC-PBB40C-40DS-0.4V-3.5-02`	0.4 mm pitch, 3.5 mm mating height, 0.3 A per contact
Headers	2.54 mm through-hole, 16 + 12 positions	0.1 in headers or sockets	Not fitted as supplied

Table 6. Connector part numbers.

Expansion boards mount back to back with the Neuro N6: the receptacles on the underside of the expansion board mate with the plugs on the underside of the Neuro N6, so the component sides of both boards face outwards. Custom expansion boards must respect the 3.5 mm mating height so that hand-soldered header pins on the Neuro N6 clear the expansion board.

6.6.1 Debug connector

PIN	SIGNAL	MCU PIN	FUNCTION
1	3V3		3.3 V rail (reference for the debug probe)
2	GND		Ground
3	SWDIO	PA13	Serial wire debug data
4	SWCLK	PA14	Serial wire debug clock
5	BOOT1	PA6	Boot mode select, shared with LCD_HSYNC. Drive high during reset for development boot
6	NRST		System reset, active low

Table 7. Debug connector pinout (M0800RS-06-GN).

The Neuro N6 Debug Adapter (OL-DEV-N6-010), available from store.neuron6.com, connects this connector to any standard debug probe, such as an ST-LINK, with the 100 mm cable supplied.

6.6.2 40-pin board-to-board connector

PIN	NET	MCU PIN	BALL	FUNCTION
1	CONN_D	PD5	D14	GPIO (expansion function D)
2	SD_D2	PC10	D8	SDMMC1 data 2 / SPI3 SCK
3	CONN_C	PE10	C14	GPIO (expansion function C)
4	I2C_SDA	PE6	A10	I2C1 SDA (3.3 V, shared with on-board sensors)
5	LCD_R4	PA7	P12	LTDC red bit 4
6	I2C_SCL	PH9	C8	I2C1 SCL (3.3 V, shared with on-board sensors)
7	LCD_G6	PB11	L10	LTDC green bit 6
8	GND			Ground
9	LCD_G7	PB10	P11	LTDC green bit 7
10	CSI_D0_P	CSI_D0P	P4	MIPI CSI-2 data lane 0 positive
11	LCD_G3	PA0	P10	LTDC green bit 3
12	CSI_D0_N	CSI_D0N	N4	MIPI CSI-2 data lane 0 negative
13	LCD_G5	PB12	M10	LTDC green bit 5
14	GND			Ground
15	CONN_F	PB4	L11	GPIO (expansion function F)
16	GND			Ground
17	LCD_B7	PA2	N8	LTDC blue bit 7
18	CSI_CLK_P	CSI_CKP	P3	MIPI CSI-2 clock lane positive

PIN	NET	MCU PIN	BALL	FUNCTION
19	LCD_B6	PA8	P8	LTDC blue bit 6
20	CSI_CLK_N	CSI_CKN	N3	MIPI CSI-2 clock lane negative
21	LCD_B4	PA10	N7	LTDC blue bit 4
22	GND			Ground
23	LCD_B3	PA11	M6	LTDC blue bit 3
24	GND			Ground
25	LCD_B5	PA9	M7	LTDC blue bit 5
26	CSI_D1_P	CSI_D1P	P2	MIPI CSI-2 data lane 1 positive
27	LCD_CLK	PA5	M8	LTDC pixel clock
28	CSI_D1_N	CSI_D1N	N2	MIPI CSI-2 data lane 1 negative
29	CONN_G	PE5	D9	GPIO (expansion function G)
30	GND			Ground
31	LCD_VSYNC	PF7	K4	LTDC vertical sync
32	GND			Ground
33	SD_D0	PC8	C6	SDMMC1 data 0
34	SD_D3	PC11	C7	SDMMC1 data 3 / SPI3 MISO
35	LCD_R6	PD13	E14	LTDC red bit 6
36	SD_CK	PC12	D7	SDMMC1 clock / SPI3 MOSI
37	LCD_R7	PD8	D13	LTDC red bit 7
38	SD_CMD	PH2	E6	SDMMC1 command
39	LCD_R3	PF4	J1	LTDC red bit 3
40	SD_D1	PC9	D6	SDMMC1 data 1

Table 8. 40-pin connector (plug) pin assignment. Expansion board receptacle pin numbers are mirrored: receptacle pin = 41 - n.

6.6.3 30-pin board-to-board connector

PIN	NET	MCU PIN	BALL	FUNCTION
1	NRST	NRST	E4	System reset, active low
2	CONN_A	PB7	H13	GPIO (expansion function A)
3	VDD1V8			1.8 V rail output
4	SAI1_FS_B	PG2	N11	SAI1 block B frame sync
5	VDD3V3			3.3 V rail output
6	LCD_G2	PA1	L9	LTDC green bit 2
7	VDD3V3			3.3 V rail output
8	GND			Ground
9	VIN			VIN rail (USB VBUS or battery, through the ideal-diode OR-ing)
10	CONN_B	PB5	M11	GPIO (expansion function B)
11	VIN			VIN rail (USB VBUS or battery, through the ideal-diode OR-ing)
12	CONN_K	PG0	M12	GPIO (expansion function K)
13	VIN			VIN rail (USB VBUS or battery, through the ideal-diode OR-ing)
14	SAI1_SD_B	PE3	G13	SAI1 block B serial data
15	VIN			VIN rail (USB VBUS or battery, through the ideal-diode OR-ing)
16	SAI1_SCK_B	PG1	M9	SAI1 block B bit clock
17	VIN			VIN rail (USB VBUS or battery, through the ideal-diode OR-ing)
18	LCD_G4	PG10	N10	LTDC green bit 4
19	LCD_DE	PG13	P9	LTDC data enable
20	DEBUG_DAT	PA13	P7	SWDIO
21	BLE_RTS	PD4	B13	USART6 RTS
22	DEBUG_CLK	PA14	P6	SWCLK
23	CONN_M	PB6	H14	GPIO (expansion function M)
24	BLE_CTS	PD3	A13	USART6 CTS
25	BLE_RX	PC7	A9	USART6 RX
26	CONN_N	PD9	H12	GPIO (expansion function N)
27	BLE_TX	PC6	A8	USART6 TX
28	LCD_R5	PA15	N12	LTDC red bit 5
29	LCD_HSYNC	PA6	L8	LTDC horizontal sync (also BOOT1)
30	SAI1_MCLK_B	PG12	N9	SAI1 block B master clock

Table 9. 30-pin connector (plug) pin assignment. Expansion board receptacle pin numbers are mirrored: receptacle pin = 30 - n for odd n and 32 - n for even n.

7 Board operation

7.1 Absolute maximum ratings

Absolute maximum ratings are stress ratings only. They are not operating conditions, and functional operation at or beyond these limits is neither implied nor guaranteed. Exposure to conditions at the limits may reduce reliability. All voltages are referenced to GND.

PARAMETER	MIN	MAX	UNIT	LIMITING ELEMENT
Voltage on USB-C VBUS or the Vin header pin	-0.3	5.5	V	Declared maximum. The AP3441 and ETA5055 regulator inputs are rated 6 V
Battery voltage, VBAT	-0.3	4.4	V	Single-cell Li-ion limit. The charger and regulators are rated 6 V or more
Voltage on any 3.3 V I/O pin (headers, connectors)	-0.3	3.6	V	STM32N6 I/O with VDDIO = 3.3 V; LSM6DSL on the I2C1 lines
Voltage on A0 to A5 while used as ADC inputs	-0.3	2.1	V	STM32N6 VDDA18ADC + 0.3 V. Measurement range 0 to 1.8 V
Current per I/O pin		20	mA	STM32N6 I/O, junction temperature up to 90 °C
Storage temperature	-40	85	°C	Component ratings

Table 10. Absolute maximum ratings.

Do not drive the I/O pins, including I2C1 on the 40-pin connector, while the board is unpowered.

> **Warning** Do not connect a second 5 V supply to the Vin header pin while USB is connected. The Vin pin is connected directly to USB VBUS, so the two supplies would be shorted together, which may damage the board or the USB host.

> **Warning** Connect only a single 3.7 V lithium-ion or lithium-polymer cell with integral protection circuitry to the battery connector. Observe polarity (pin 1 positive), never charge unattended and do not pull on the battery lead. Any other cell or battery may cause permanent damage, fire or injury.

7.2 Recommended operating conditions

PARAMETER	MIN	MAX	UNIT	CONDITIONS
Ambient temperature	0	30	°C	Indoor, dry locations only
Relative humidity	25	75	%	Non-condensing
USB-C supply voltage	4.5	5.5	V	5 V DC via USB-C

Table 11. Recommended operating conditions.

7.3 Electrical characteristics

PARAMETER	MIN	TYP	MAX	UNIT	CONDITIONS
USB-C supply voltage (VBUS)	4.5	5.0	5.5	V	
Battery voltage, VBAT	3.5	3.7	4.24	V	Charger float voltage 4.2 V typical
Battery charge current	465	500	535	mA	TP4054, 2 kΩ PROG resistor, constant-current phase. Reduced by the charger thermal regulation when the battery voltage is low
Supply current		150 to 500		mA	Typical, depending on the workload and the expansion board fitted
3.3 V rail (VDD_3V3)	3.23	3.33	3.44	V	AP3441 buck converter, 3 A
1.8 V rail (VDD_1V8)	1.75	1.8	1.85	V	AP3441 buck converter, 3 A
1.8 V analog rail (VDDA_1V8, AREF)	1.75	1.8	1.85	V	ETA5055 LDO from VDD_3V3, including load regulation
Digital input low level, VIL			1.05	V	STM32N6 I/O, VDDIO = 3.3 V
Digital input high level, VIH	1.61			V	STM32N6 I/O, VDDIO = 3.3 V
Digital output high level, VOH	2.9			V	IIO = 6.5 mA (speed 00) to 20 mA (speed 11)
Digital output low level, VOL			0.4	V	IIO = 6.5 mA (speed 00) to 20 mA (speed 11)
ADC input range (A0 to A5)	0		1.8	V	12-bit ADC, VREF+ = 1.8 V, up to 4.67 Msps

Table 12. Electrical characteristics at 25 °C unless stated.

> **Caution** Hot surfaces. Components can become hot during sustained AI, camera or radio workloads, and the charger can become hot while charging. Do not hold the board while it is running; allow it to cool before handling.

7.4 Power supply

The Neuro N6 is powered from 5 V on the USB-C port or from a single-cell 3.7 V Li-ion battery on the battery connector. The Vin header pin is connected directly to USB VBUS. USB and battery may be connected at the same time: the TP4054 charger then charges the battery at up to 500 mA, and two ideal-diode circuits (DZDH0401DW controllers driving PMPB15XP MOSFETs) select the higher of VBUS and VBAT to form the VIN rail without diode losses. VIN supplies the regulators and the VIN contacts of the 30-pin connector, so it follows the battery voltage (3.5 to 4.2 V) when USB is not connected.

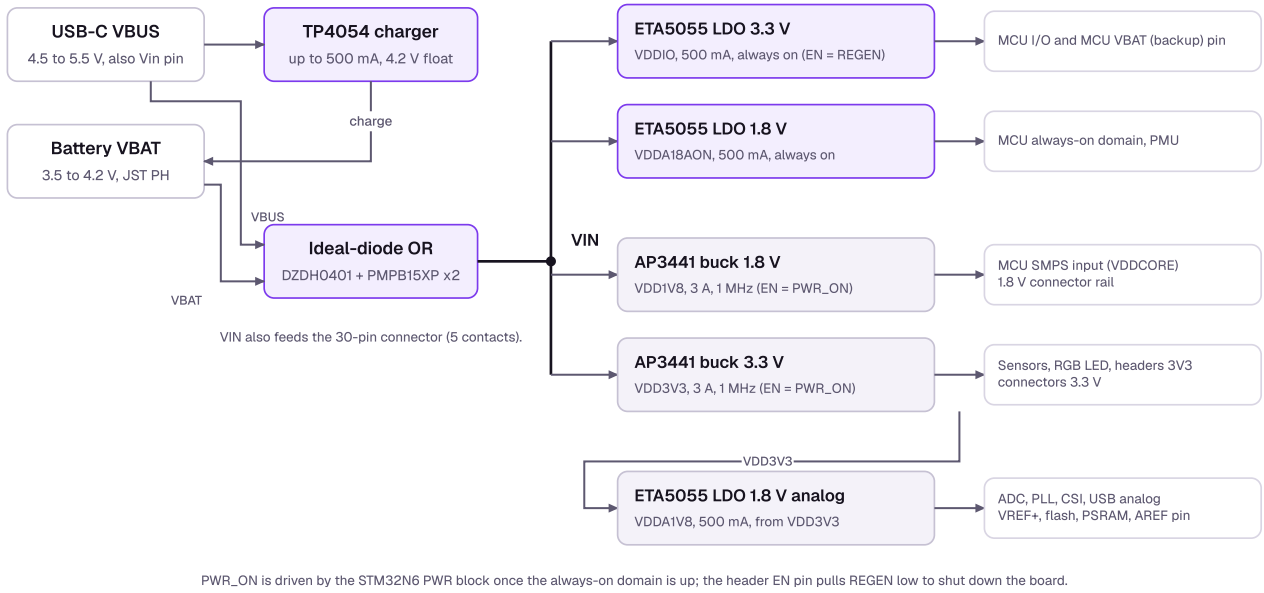


Figure 6. Neuro N6 power tree.

Two always-on ETA5055 regulators generate the 3.3 V VDDIO supply for the microcontroller I/O and the 1.8 V always-on domain. VDDIO also supplies the microcontroller VBAT pin, so the real-time clock has no separate backup supply and keeps time only while the board is powered. They are enabled by the REGEN net, which is pulled up to VIN and brought out on the EN header pin. Once the always-on domain is up, the STM32N6 asserts PWR_ON, which enables the two AP3441 buck converters (1.8 V and 3.3 V) and, from the 3.3 V rail, the ETA5055 analog regulator that produces VDDA_1V8 for the ADC, PLLs, CSI D-PHY, USB PHY, external memories and the AREF pin.

> **Note** Battery: use a protected single cell with a JST PH plug wired pin 1 positive; check the polarity, as it differs between suppliers. The charger is programmed to 500 mA, so use a cell rated for that charge current. The board does not monitor the battery temperature: charge only between 0 and 45 °C. Charging shares the USB supply with the board, so charging and heavy processing together draw more current from the USB port.

7.5 Power rail outputs

External circuits can draw power from the rails below. Check that the source connected to the Neuro N6 (USB-C or battery) can supply the whole system.

RAIL	NOMINAL	AVAILABLE ON	MAXIMUM EXTERNAL LOAD	LIMITED BY
`VBUS`	5 V (USB only)	Vin header pin	Depends on the USB source	USB source, shared with the board and the charger; not powered from the battery
`VIN`	USB VBUS or VBAT (3.5 to 4.2 V)	30-pin connector (5 contacts)	1.5 A	0.3 A per connector contact
`VDD_3V3`	3.3 V	3V3 header pin, 30-pin connector (2 contacts)	0.6 A (connector)	Connector contacts; AP3441 3 A shared with the board
`VDD_1V8`	1.8 V	30-pin connector (1 contact)	0.3 A	Connector contact rating
`VDDA_1V8`	1.8 V	AREF header pin	10 mA	Reference use only: small ETA5055 regulator that also supplies the ADC reference and the memories

Table 13. Power rail outputs.

> **Caution** When powering external devices from the rails, keep each rail within the limits above. Overloading a rail causes voltage droop or brownout and can damage devices on the affected rail.

7.6 Boot modes

The STM32N6 boot ROM selects the boot mode from its BOOT0 and BOOT1 pins. On the Neuro N6, BOOT0 is not connected and BOOT1 is available on the debug connector, which gives two hardware boot modes. The Ohm Lab first-stage bootloader (FSBL) then provides two software modes.

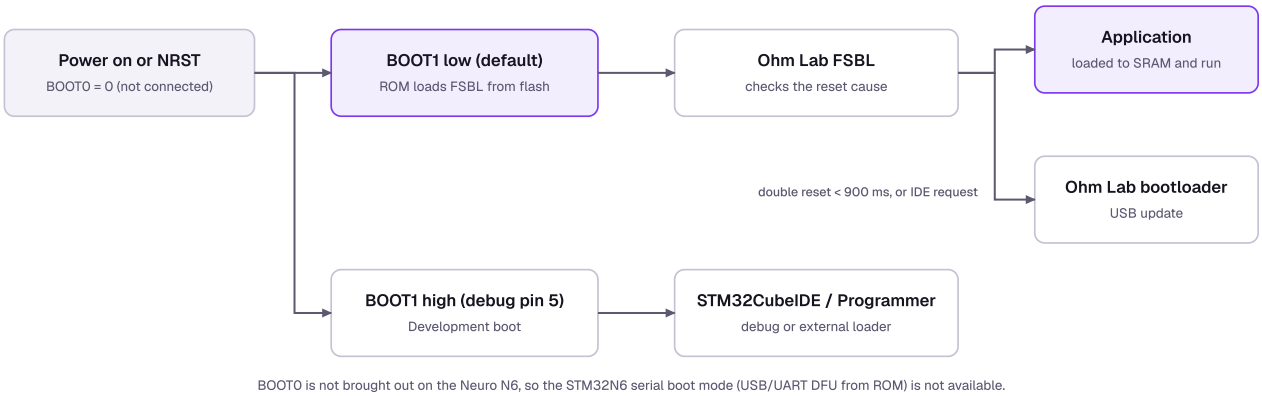


Figure 7. Boot sequence.

7.6.1 Boot from flash

Default mode. The boot ROM loads the Ohm Lab FSBL from the external flash, and the FSBL copies the application into internal SRAM and runs it. The maximum application size is 1.5 MB (1,572,832 bytes), set by the internal SRAM available for the copied application. It requires Ohm Lab bootloader v2.4.0 or later; the current Arduino core requires v2.6.1 or later.

7.6.2 Ohm Lab bootloader

The application firmware and neural network weights are updated through the Ohm Lab bootloader over the USB-C port, from the Arduino IDE or the Neuro N6 Uploader. The bootloader is entered in two ways:

- **Automatic entry**: applications built with the Arduino core include a mandatory routine that detects the IDE upload request and resets into the bootloader.
- **Manual entry**: press RESET twice with less than 900 ms between presses. This route needs no application support and recovers a board with a corrupted application.

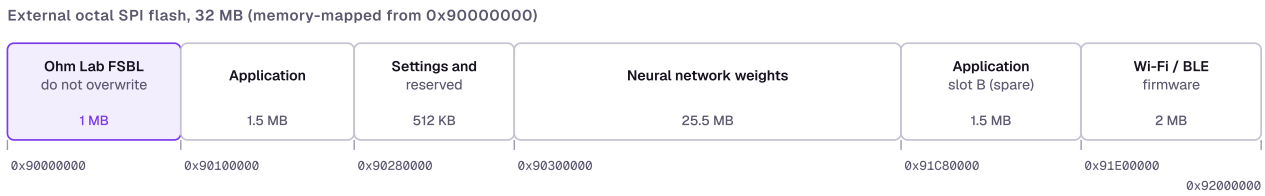


Figure 8. External flash memory map. Segment widths are not to scale. The software wiki at wiki.neuron6.com describes the regions in detail.

> **Caution** When programming the external flash directly, do not overwrite the address range 0x90000000 to 0x90100000. This region holds the Ohm Lab FSBL, which controls normal boot and the bootloader. If it is corrupted the board can be recovered with the development boot mode below.

7.6.3 Development boot

Driving BOOT1 (debug connector pin 5) high during reset selects the STM32N6 development boot mode. The board can then be debugged with STM32CubeIDE or programmed with STM32CubeProgrammer using the Neuro N6 external memory loader. This mode recovers a board on which the FSBL has been erased or corrupted. Connect any standard debug probe, such as an ST-LINK, through the Neuro N6 Debug Adapter (OL-DEV-N6-010), available from store.neuron6.com.

7.6.4 Serial boot

The STM32N6 serial boot mode (USB and UART DFU from the boot ROM) requires BOOT0 high. BOOT0 is not brought out on the Neuro N6, so this mode is not available.

For more information see the STM32N6 datasheet, reference manual RM0484 and user manual UM3234 (boot ROM), and the Ohm Lab FSBL documentation at wiki.neuron6.com.

8 Peripheral interfaces

This section lists the STM32N6 peripherals exposed on the Neuro N6 with their MCU pins, balls and connector pins. Connector pins are given as `40DP:n` and `30DP:n` (plug pin numbers on the Neuro N6) and `HDR:name` for the headers. Refer to the STM32N6 datasheet and reference manual for peripheral details.

8.1 MIPI CSI-2

One 2-lane MIPI CSI-2 receiver with internal D-PHY, compliant with CSI-2 version 1.3 and up to 2.5 Gbit/s per lane. The lanes pass through EMI8141MUTAG common-mode filters with ESD protection before the connector.

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
`CSI_D0_P`	`CSI_D0P`	`P4`	`40DP:10`	Data lane 0 positive
`CSI_D0_N`	`CSI_D0N`	`N4`	`40DP:12`	Data lane 0 negative
`CSI_CLK_P`	`CSI_CKP`	`P3`	`40DP:18`	Clock lane positive
`CSI_CLK_N`	`CSI_CKN`	`N3`	`40DP:20`	Clock lane negative
`CSI_D1_P`	`CSI_D1P`	`P2`	`40DP:26`	Data lane 1 positive
`CSI_D1_N`	`CSI_D1N`	`N2`	`40DP:28`	Data lane 1 negative

Table 14. MIPI CSI-2 pins.

8.2 SDMMC1

One 4-bit SDMMC interface (SDMMC1) for an SD card or a Wi-Fi module, at clock rates up to 50 MHz (3.3 V signalling, SD high speed). Three of the pins can be used as SPI3 instead.

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
`SD_CMD`	`PH2`	`E6`	`40DP:38`	Command
`SD_CK`	`PC12`	`D7`	`40DP:36`	Clock (SPI3_MOSI)
`SD_D0`	`PC8`	`C6`	`40DP:33`	Data 0
`SD_D1`	`PC9`	`D6`	`40DP:40`	Data 1
`SD_D2`	`PC10`	`D8`	`40DP:2`	Data 2 (SPI3_SCK)
`SD_D3`	`PC11`	`C7`	`40DP:34`	Data 3 (SPI3_MISO)

Table 15. SDMMC1 pins.

8.3 I2C

Two I2C interfaces are available, each with 4.7 kΩ pull-ups to 3.3 V. I2C1 serves the on-board IMU and magnetometer and the 40-pin connector, so the sensors of the Neuro Vision boards share this bus. I2C2 is dedicated to the header pins. Both support standard mode (100 kHz) and fast mode (400 kHz).

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
`I2C_SCL`	`PH9`	`C8`	`40DP:6`	I2C1 clock
`I2C_SDA`	`PE6`	`A10`	`40DP:4`	I2C1 data
`SCL`	`PD14`	`D12`	`HDR:SCL`	I2C2 clock
`SDA`	`PD15`	`C12`	`HDR:SDA`	I2C2 data

Table 16. I2C pins.

DEVICE	BUS	7-BIT ADDRESS
LSM6DSL	I2C1	`0x6A`
MMC5603NJ	I2C1	`0x30`
OV5640 (Neuro Vision OV5640, TFT)	I2C1	`0x3C`
ST camera promodule, VD66GY as standard (Neuro Vision ST Cam, Thermal)	I2C1	`0x10`
VL53L1CB (Neuro Vision OV5640, ST Cam, TFT)	I2C1	`0x29`
Lepton 3.1R (Neuro Vision Thermal)	I2C1	`0x2A`
GT911 touch controller (Neuro Vision TFT)	I2C1	`0x5D` or `0x14`

Table 17. I2C addresses on the platform.

8.4 SPI

SPI2 is available on the headers. SPI3 is available as an alternate function on three SDMMC1 pins of the 40-pin connector. The Neuro Vision Thermal board reads its thermal imager on SPI1 (SCK PA5, MISO PB4, chip select PF4).

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
`SCK`	`PF2`	`K1`	`HDR:SCK`	SPI2 clock
`MISO`	`PD11`	`F11`	`HDR:MISO`	SPI2 MISO
`MOSI`	`PD7`	`B11`	`HDR:MOSI`	SPI2 MOSI
`SD_D2`	`PC10`	`D8`	`40DP:2`	SPI3 clock (shared with SDMMC1)
`SD_D3`	`PC11`	`C7`	`40DP:34`	SPI3 MISO (shared with SDMMC1)
`SD_CK`	`PC12`	`D7`	`40DP:36`	SPI3 MOSI (shared with SDMMC1)

Table 18. SPI pins.

8.5 UART

UART4 is available on the headers as a two-wire interface. USART6 is a four-wire interface with hardware flow control on the 30-pin connector, used for the BLE HCI link on the wireless expansion boards.

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
RX	PD0	A12	HDR:RX	UART4 receive
TX	PD1	B12	HDR:TX	UART4 transmit
BLE_RX	PC7	A9	30DP:25	USART6 receive
BLE_TX	PC6	A8	30DP:27	USART6 transmit
BLE_RTS	PD4	B13	30DP:21	USART6 request to send
BLE_CTS	PD3	A13	30DP:24	USART6 clear to send

Table 19. UART pins.

8.6 SAI

Block B of the SAI1 serial audio interface is exposed on the 30-pin connector for I2S audio input or output.

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
SAI1_MCLK_B	PG12	N9	30DP:30	Master clock
SAI1_SCK_B	PG1	M9	30DP:16	Bit clock
SAI1_FS_B	PG2	N11	30DP:4	Frame sync
SAI1_SD_B	PE3	G13	30DP:14	Serial data

Table 20. SAI1 block B pins.

8.7 GPIO and analog inputs

Seven digital I/O and six analog inputs are available on the headers, and every pin can be used as a 3.3 V GPIO. The analog inputs connect to the 12-bit ADC1 and ADC2 converters, whose full-scale input is 1.8 V.

NET	MCU PIN	BALL	CONNECTOR PIN	FUNCTION
D5	PE9	B14	HDR:D5	Digital I/O, TIM1_CH1 PWM
D6	PF5	J4	HDR:D6	Digital I/O
D9	PD12	E11	HDR:D9	Digital I/O
D10	PF10	J3	HDR:D10	Digital I/O
D11	PF15	K3	HDR:D11	Digital I/O
D12	PA3	K11	HDR:D12	Digital I/O, TIM16_CH1 PWM
D13	PG14	L5	HDR:D13	Digital I/O, TIM8_CH4 PWM
A0	PF13	M2	HDR:A0	ADC2_INP2
A1	PF11	L2	HDR:A1	ADC1_INP2
A2	PF12	L3	HDR:A2	ADC1_INP6
A3	PF3	J5	HDR:A3	ADC1_INP16
A4	PF14	K2	HDR:A4	ADC2_INP6, TIM2_CH2 PWM
A5	PG15	M5	HDR:A5	ADC1_INP7 and ADC2_INP7, TIM1_CH4 PWM

Table 21. Header GPIO and analog inputs.

> **Note** The ADC reference (VREF+) is the 1.8 V VDDA_1V8 rail. An analog input above 1.8 V reads full scale and must not exceed 1.8 V while its ADC channel is in use. Use a resistive divider to measure 3.3 V or 5 V signals.

NET	MCU PIN	BALL	CONNECTOR PIN	FUNCTION
VBATSEN	PF6	J2	internal	Battery voltage sense, ADC1_INP15. Divider 100 kΩ / 56 kΩ: 4.2 V reads 1.51 V
VBSEN	PB0	G14	internal	USB VBUS present, digital input through a 100 kΩ / 100 kΩ divider
USER	PG8	L12	internal	USER button, active high
IMU_INT1	PE14	C9	internal	LSM6DSL interrupt 1
IMU_INT2	PE15	B10	internal	LSM6DSL interrupt 2
PWR_ON	PWR_ON	E3	internal	Enables the buck converters and the analog regulator

Table 22. Internal monitoring and control signals.

8.8 RMII Ethernet

The header pins can be configured as a reduced media independent interface (RMII) to an external 10/100 Mbit/s Ethernet PHY, for example on a custom expansion board, giving Ethernet conforming to IEEE 802.3. In this mode the listed header functions are unavailable.

NET	MCU PIN	BALL	CONNECTOR PIN	DESCRIPTION
A5	PG15	M5	HDR:A5	ETH1_RMII_REF_CLK, 50 MHz reference clock
D6	PF5	J4	HDR:D6	ETH1_CLK
D10	PF10	J3	HDR:D10	ETH1_RMII_CRS_DV
A4	PF14	K2	HDR:A4	ETH1_RMII_RXD0
D11	PF15	K3	HDR:D11	ETH1_RMII_RXD1
A1	PF11	L2	HDR:A1	ETH1_RMII_TX_EN
A2	PF12	L3	HDR:A2	ETH1_RMII_TXD0
A0	PF13	M2	HDR:A0	ETH1_RMII_TXD1
D9	PD12	E11	HDR:D9	ETH1_MDIO
TX	PD1	B12	HDR:TX	ETH1_MDC

Table 23. RMII pin assignment.

8.9 LTDC display interface

The LCD-TFT display controller is exposed in 16-bit RGB565 mode on the 40-pin and 30-pin connectors for the Neuro Vision TFT boards. The controller supports resolutions up to XGA. The pin assignment is given in the connector tables in the hardware interfaces section: red bits 3 to 7, green bits 2 to 7 and blue bits 3 to 7 with LCD_CLK, LCD_DE, LCD_HSYNC and LCD_VSYNC.

8.10 USB

The USB-C port connects to the OTG1 high-speed instance of the STM32N6 through a USBLC6-2P6 ESD protection device. The port operates as a USB 2.0 high-speed device at 480 Mbit/s. The CC1 and CC2 lines carry 5.1 kΩ pull-down resistors, so the board identifies itself as a 5 V sink to a USB-C source.

9 Regulatory compliance

This section describes the regulatory status of the Neuro N6 as supplied by Ohm Lab. The Neuro N6 is a development board intended for evaluation and development use, and is assessed together with its expansion boards as a platform. Where the board is incorporated into a finished product, responsibility for the compliance of that product rests with its manufacturer.

9.1 Product marking

The following markings apply to the Neuro N6 and to its packaging. The controlled artwork is held in the technical file.

SYMBOL	MARKING	APPLIED TO	MEANING
	CE	Board and packaging	Conformity with the EMC Directive 2014/30/EU and the RoHS Directive 2011/65/EU.
	Crossed-out wheeled bin	Board and packaging	Do not dispose of with unsorted household waste. The bar identifies equipment placed on the market after 13 August 2005.
	Triman and Info-tri	Packaging	France: sorting information required by the French environment code. The Triman is printed on the carton; an Info-tri sticker with the sorting instructions is added to orders for France.

Table 24. Product markings.

No UKCA marking is applied: CE marking is accepted for goods placed on the Great Britain market. RoHS conformity is declared through the CE marking and carries no separate mark.

9.2 Electromagnetic compatibility

The Neuro N6 and its Neuro Vision expansion boards have been assessed for electromagnetic compatibility as a product family, based on tests of worst-case configurations of the Neuro N6 with expansion boards fitted. The Neuro N6 conforms to the EMC Directive 2014/30/EU, with emissions assessed to EN 55032 Class B and immunity to EN 55035. The assessment applies to the Neuro N6 operated with the default clock and radio settings of the Neuro N6 Arduino core. The declaration of conformity is available from Ohm Lab on request.

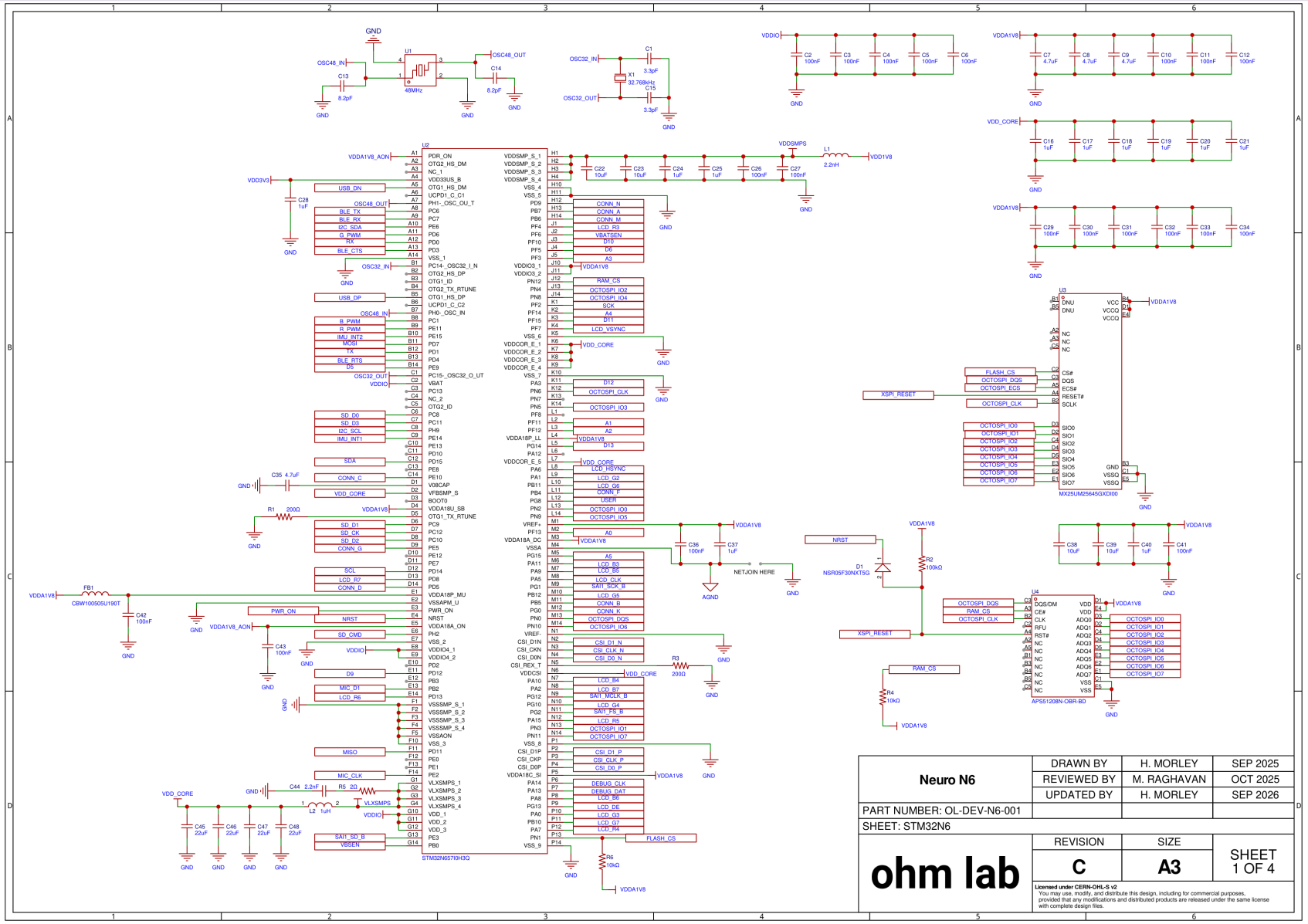
9.3 Materials and disposal

The Neuro N6 is manufactured to the substance restrictions of Directive 2011/65/EU (RoHS 2) as amended and the UK Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012. Supplier material declarations, and declarations covering substances of very high concern under REACH, are held in the technical file.

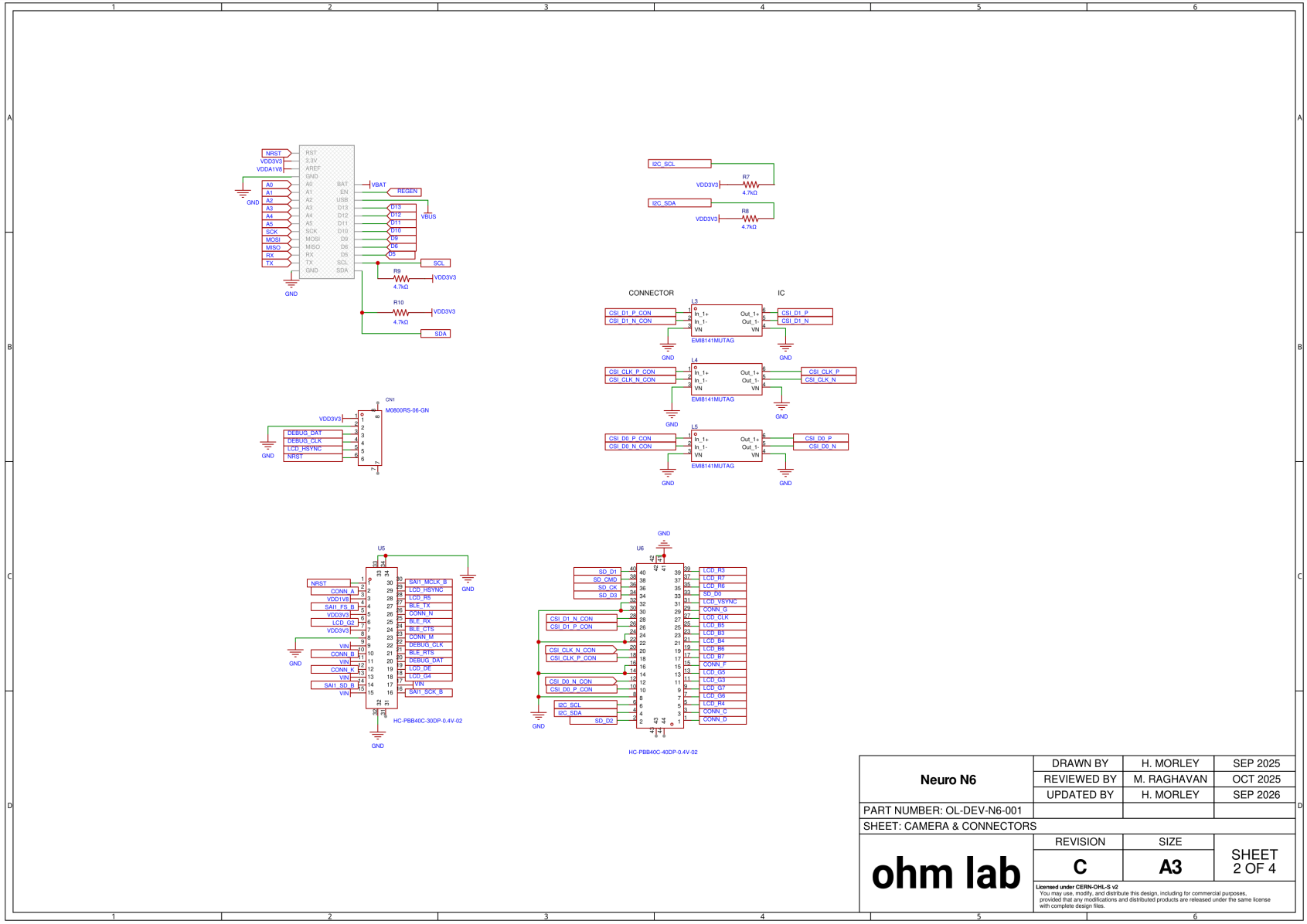
At end of life, do not dispose of the board with unsorted household waste. Return it to a designated collection point for electrical and electronic equipment, or to Ohm Lab.

10 Schematics

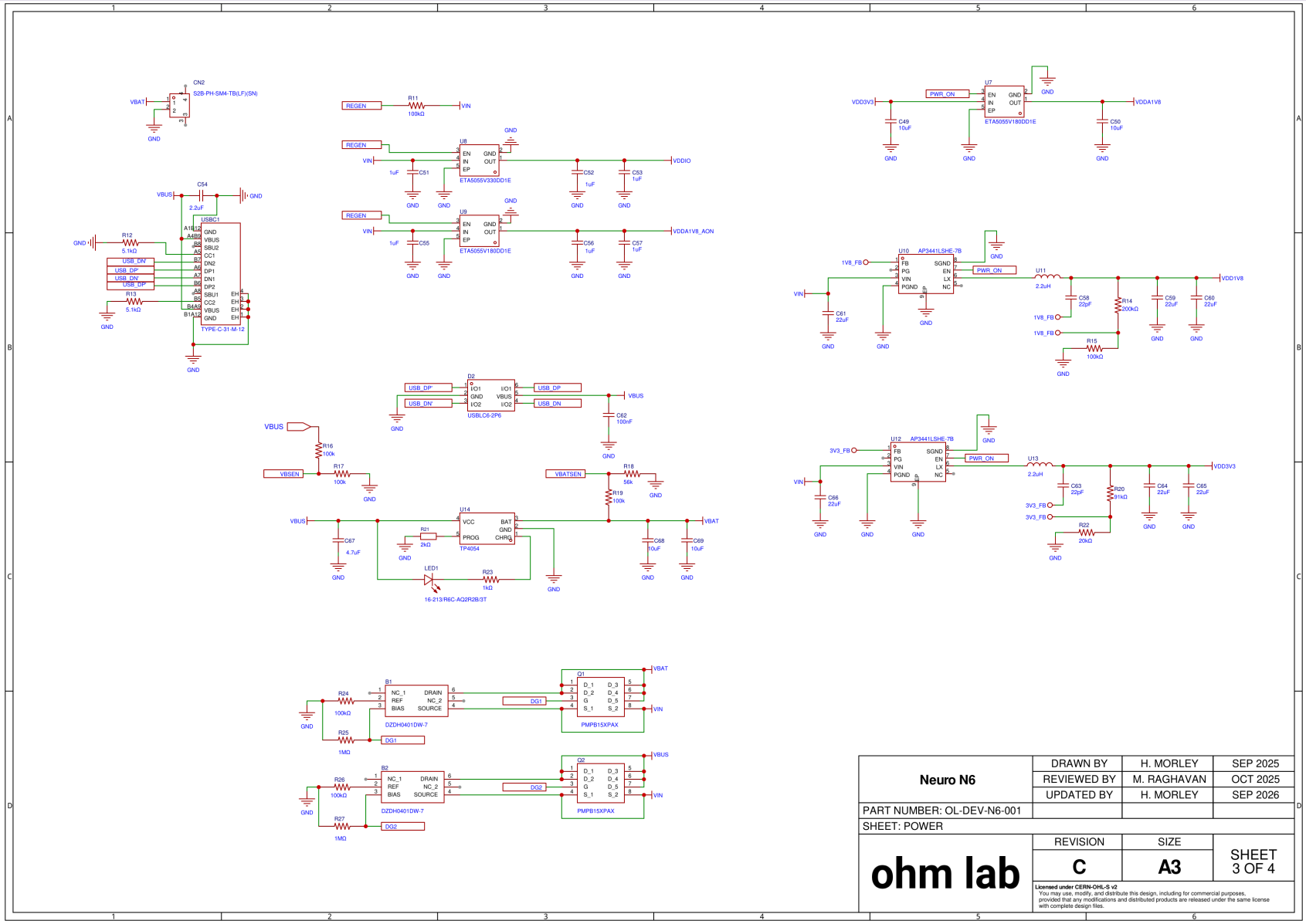
The Rev C schematics are reproduced on the following pages. Net names match those used in this document; on the connector side of the common-mode filters the CSI-2 nets carry a _CON suffix. The schematics and mechanical files are published in the Neuro N6 Hardware repository (github.com/ohmlab-ltd/Neuro-N6-Hardware) under the CERN-OHL-S v2 licence.



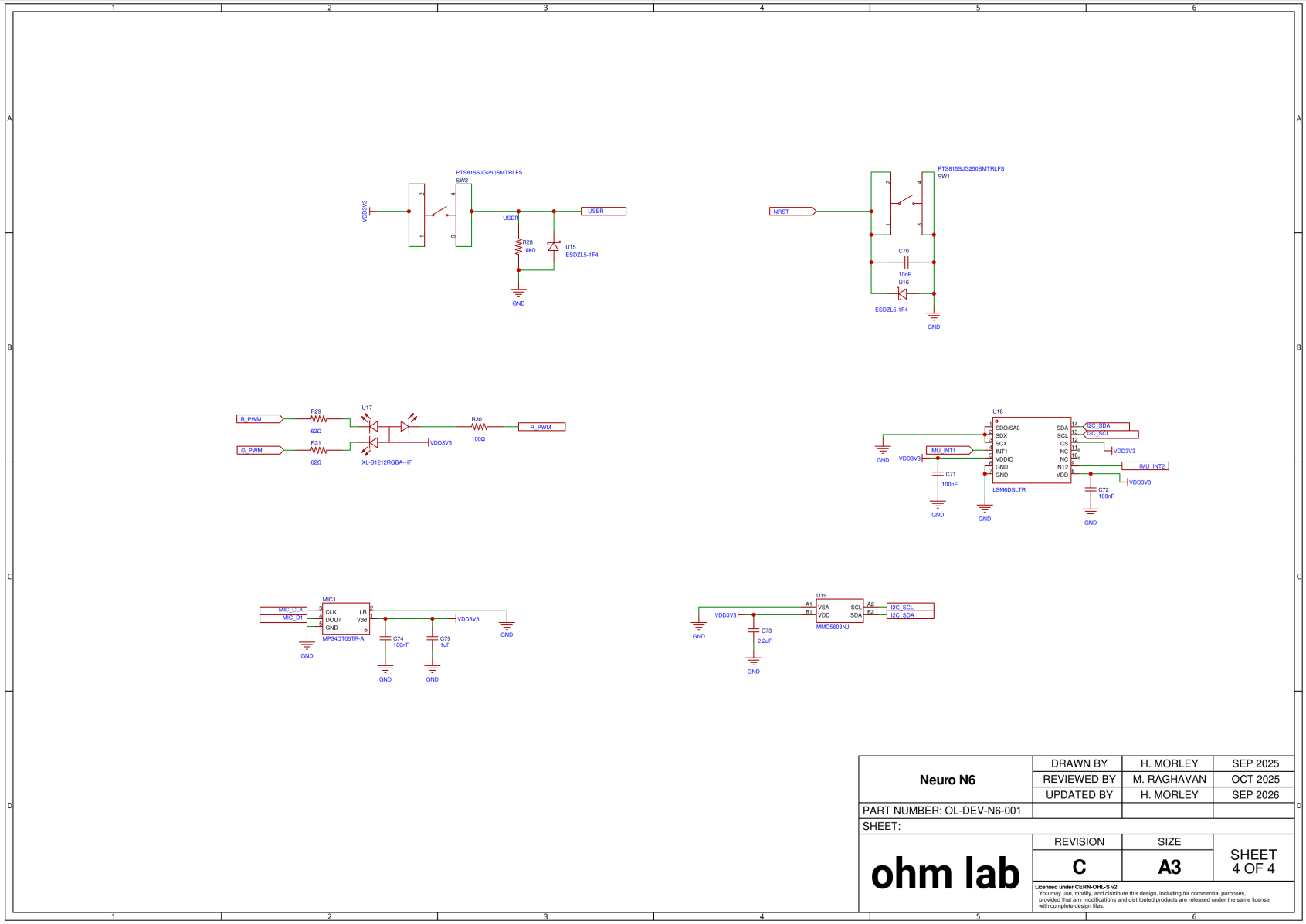
Neuro N6 Rev C schematic, sheet 1 of 4: STM32N6.



Neuro N6 Rev C schematic, sheet 2 of 4: camera interface and connectors.



Neuro N6 Rev C schematic, sheet 3 of 4: power.



Neuro N6 Rev C schematic, sheet 4 of 4: sensors and user interface.

11 Document revision history

REVISION	DATE	CHANGES
α	15/03/2026	• Pre-production first release
α.01	21/03/2026	• 32 MB PSRAM option removed
B	01/07/2026	• Production release
C	02/09/2026	• Ordering information, ratings and regulatory section added
D	24/09/2026	• Rev C hardware • Pinouts, power tree and boot sequence added
E	25/10/2026	• Vin pin documented as USB VBUS • Operating conditions and ratings corrected • Flash memory map added

Table 25. Document revision history.